

PDDL: Torri di Hanoi

1.1 Modellazione del Dominio

Predicati

$$P = \{\text{Asse}/1, \text{Disco}/1, \text{smaller}/2, \text{free}/1, \text{diskOn}/2\}$$

Azioni

- **Sposta**(x, y', y)
 - *Pre* : $\text{Disco}(x) \wedge \text{free}(x) \wedge \text{free}(y) \wedge \text{diskOn}(x, y') \wedge (y \neq y') \wedge \text{smaller}(x, y)$
 - *Eff* : $\neg \text{free}(y) \wedge \text{diskOn}(x, y) \wedge \neg \text{diskOn}(x, y') \wedge \text{free}(y')$

1.2 Istanziazione specifica

Istanziazione (Oggetti)

$$F = \{d_1, d_2, d_3, d_4, A, B, C\}$$

Stato Iniziale

$$\begin{aligned} I = & \text{Asse}(A) \wedge \text{Asse}(B) \wedge \text{Asse}(C) \wedge \\ & \text{Disco}(d_1) \wedge \text{Disco}(d_2) \wedge \text{Disco}(d_3) \wedge \text{Disco}(d_4) \wedge \\ & \text{smaller}(d_1, d_2) \wedge \text{smaller}(d_1, d_3) \wedge \text{smaller}(d_1, d_4) \wedge \\ & \text{smaller}(d_2, d_3) \wedge \text{smaller}(d_2, d_4) \wedge \text{smaller}(d_3, d_4) \wedge \\ & \text{smaller}(d_1, A) \wedge \text{smaller}(d_2, A) \wedge \text{smaller}(d_3, A) \wedge \text{smaller}(d_4, A) \wedge \\ & \text{smaller}(d_1, B) \wedge \text{smaller}(d_2, B) \wedge \text{smaller}(d_3, B) \wedge \text{smaller}(d_4, B) \wedge \\ & \text{smaller}(d_1, C) \wedge \text{smaller}(d_2, C) \wedge \text{smaller}(d_3, C) \wedge \text{smaller}(d_4, C) \wedge \\ & \text{free}(d_1) \wedge \text{free}(B) \wedge \text{free}(C) \wedge \\ & \text{diskOn}(d_4, A) \wedge \text{diskOn}(d_3, d_4) \wedge \text{diskOn}(d_2, d_3) \wedge \text{diskOn}(d_1, d_2) \end{aligned}$$

Obiettivo

$$G = \text{diskOn}(d_4, B) \wedge \text{diskOn}(d_3, d_4) \wedge \text{diskOn}(d_2, d_3) \wedge \text{diskOn}(d_1, d_2)$$